

PREPARED FOR SUBMISSION TO JINST

TOPICAL WORKSHOP ON ELECTRONICS FOR PARTICLE PHYSICS - TWEPP2022

19 - 23 SEPTEMBER, 2022

BERGEN, NORWAY

Radiation studies performed on the High Luminosity ATLAS TileCal link Daughterboard

E. Valdes Santurio^{a,1} , S. Silverstein^a , C. Bohm^a , H. Motzkau^a , C. Clement^a , K. Dunne^a , S. Lee^a on behalf of the ATLAS Tile Calorimeter System^{2,3}

^a*Stockholm University, Stockholm, Sweden*

E-mail: eduardo.valdes@fysik.su.se

ABSTRACT: The new electronics of the ATLAS Tile Calorimeter for the HL-LHC interfaces the on-detector and off-detector electronics by means of a Daughterboard. The Daughterboard is positioned on-detector featuring commercial SFPs+, CERN GBTx ASICs, ProASIC FPGAs and Kintex Ultrascale FPGAs. The design minimizes single points of failure and mitigates radiation damage by means of a double redundant scheme, Triple Mode Redundancy, Xilinx Soft Error Mitigation IP, CRC/FEC for link data transfer, and SEL protection circuitry. We present an updated summary of the TID, NIEL and SEE qualification tests, and performance studies of the Daughterboard revision 6 design.

KEYWORDS: Radiation damage to electronic components, Front-end electronics for detector readout.

¹Corresponding author.

²This work was supported by Stockholm University and CERN.

³Copyright 2020, CERN, for the benefit of the ATLAS Collaboration. CC-BY-4.0 license.

